IB MATH AA SL

Lesson 3

Geometric Sequences & Series

Number & Algebra · 60-minute lesson

Syllabus SL 1.3 ·

Do Now — recall

ON YOUR WHITEBOARD

1. In 2, 6, 18, 54, … what is multiplied each time?
2. Write the next term of 3, 6, 12, …
3. How is this different from an arithmetic sequence?

Recall:

$$u_n = u_1 + (n-1)d$$

Answers (click to reveal)

1. × 3 each time
2. 24
3. Geometric MULTIPLIES by r; arithmetic ADDS d

What we're learning today

LEARNING OBJECTIVES

1. **Recognise**
   a geometric sequence and find the common ratio r.

2. **Apply**
   $u_n = u_1 r^{n-1}$ to find any term.

3. **Apply**
   $S_n = u_1(r^n - 1)/(r - 1)$ to sum a finite series.

4. **Solve**
   growth and decay problems such as compound interest.

SUCCESS CRITERIA — I CAN...

- ✓ find r by dividing a term by the one before it
- ✓ use $u_n = u_1 r^{n-1}$ to find any term
- ✓ choose the right S_n formula and evaluate it
- ✓ model a real situation with a geometric sequence

The nth term

THE FORMULA

$$u_n = u_1 \; r^{\,n-1}$$

u_1
first term

r
common ratio = $u_{n+1} \div u_n$

n
term number

WHY (n−1)

$u_1 = u_1$

$u_2 = u_1 r$

$u_3 = u_1 r^2$

$u_n = u_1 r^{n-1}$ ← one fewer power than n

Find a term

QUESTION

A geometric sequence has $u_1 = 5$ and $r = 3$.

Find the 7th term.

SOLUTION

Step 1 $u_1 = 5$, $r = 3$, $n = 7$

Step 2 substitute:

$$u_7 = 5 \times 3^{7-1} = 5 \times 3^6$$

Step 3 evaluate ($3^6 = 729$):

$$u_7 = 5 \times 729 = 3645$$

Tip:

Substitute before you evaluate — method marks live on the substitution line.

Your turn — nth term

ANSWER IN YOUR BOOK

1. Find the 5th term of 3, 12, 48, …
2. A GP has $u_1 = 80$, $r = ½$. Find u_4.
3. Find r for 100, 20, 4, …

Answers (click to reveal)

1. $3·4^4 = 768$
2. $80·(½)^3 = 10$
3. $r = 20/100 = 0.2$

The sum of n terms

TWO FORMS (same thing)

$$S_n = \frac{u_1(r^n - 1)}{r - 1}$$

$$S_n = \frac{u_1(1 - r^n)}{1 - r}$$

Top
use when $r > 1$

Bottom
use when $r < 1$ (keeps it positive)

GDC (Paper 2)

menu → Sequence / Σ

or Σ(u_1·r^(x−1), x, 1, n)

Paper 1:
show the formula and working.

Sum a series

QUESTION

Find the sum of the first 8 terms of

$3 + 6 + 12 + \ldots$

SOLUTION

Step 1 $u_1 = 3$, $r = 2$, $n = 8$ $(r > 1)$

Step 2 substitute:

$$S_8 = \frac{3(2^8 - 1)}{2 - 1}$$

Step 3 evaluate ($2^8 = 256$):

$$S_8 = \frac{3(255)}{1} = 765$$

Your turn — sums

ANSWER IN YOUR BOOK

1. Sum the first 6 terms of $4 + 12 + 36 + \ldots$

2. Sum the first 10 terms of $1 + 2 + 4 + \ldots$

3. £2000 at 4%/yr — value after 10 years?

Answers (click to reveal)

1. $4(3^6-1)/2 = 1456$
2. $1(2^{10}-1)/1 = 1023$
3. $2000(1.04)^{10} \approx$ £2960.49

Diagnostic check

A student writes: "For 2, 6, 18, ..., the 5th term is $u_5 = 2 \times 3^5 = 486$."

Is the student right? If not, what is the mistake and the correct answer?

Wrong — the exponent should be $(n-1)$, not n. Correct: $u_5 = 2 \times 3^4 = 162$. Using 3^5 counts one ratio too many.

EXAM-STYLE QUESTION

Exam-style question

The third term of a geometric sequence is 20 and the sixth term is 160.

(a) Show that the common ratio $r = 2$. [3]

(b) Find the first term u_1. [2]

(c) Hence find the sum of the first 10 terms. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $r^3 = 160 \div 20 = 8$ (M1)(A1) $\to r = 2$ (AG)

(b) $u_1(2^2) = 20$ (M1) $\to u_1 = 5$ (A1)

(c) $S_{10} = 5(2^{10}-1)/(2-1)$ (M1)(A1) $= 5115$ (A1)

Common mistakes

✕ **(n−1), not n**

$u_5 = u_1r^4$, not u_1r^5.

✕ **Wrong S_n form**

If $r < 1$, use $(1-r^n)/(1-r)$ to avoid sign errors.

✕ **Ratio direction**

r = later ÷ earlier, e.g. $u_2 \div u_1$.

✕ **Rounding early**

Keep exact values; round only the final answer (3 s.f.).

Mixed practice

QUESTIONS

1. Find the 9th term of 2, 6, 18, …

2. Sum the first 6 terms of 5 + 10 + 20 + …

3. A GP has 2nd term 12 and 5th term 96. Find r and u_1.

4. £2000 at 4%/yr — value after 10 years?

Answers (click to reveal)

1. $2 \cdot 3^8 = 13\,122$
2. $5(2^6 - 1)/1 = 315$
3. $r^3 = 96/12 = 8 \rightarrow r = 2$; $u_1 = 6$
4. $2000(1.04)^{10} \approx$ £2960.49

Plenary & exit ticket

KEY FORMULAS

$$u_n = u_1 \; r^{\,n-1}$$

$$S_n = \frac{u_1(r^n - 1)}{r - 1}$$

EXIT TICKET — before you leave

1. Find the 6th term of 4, 8, 16, …
2. Find r for 90, 30, 10, …
3. Sum the first 5 terms of 1 + 3 + 9 + …

Answers (click to reveal)

1. $4 \cdot 2^5 = 128$
2. $r = 1/3$
3. $1(3^5 - 1)/2 = 121$

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Geometric sequences & series.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 4

Sum to infinity — what happens to the sum when $|r| < 1$ and the terms shrink.

$$S_\infty = \frac{u_1}{1 - r}$$